

SIBI K

Health Data Scientist | Data Analytics | Machine Learning

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PROFESSIONAL SUMMARY

M.Sc. Health Data Science candidate with a clinical background in Anaesthesia and Operation Technology and hands-on experience in healthcare data analytics, data engineering, machine learning, and deep learning. Skilled in Python, SQL, and Power BI, with practical experience working on real-world clinical datasets, longitudinal patient records, and healthcare ETL pipelines. Experienced in building data-cleaning workflows, predictive models, dashboards, and production-oriented analytics solutions.

CORE SKILLS

Programming & Analytics: Python, SQL, Pandas, NumPy, Data Cleaning, EDA, Data Analysis

BI & Statistics: Power BI, Plotly, Dash, Biostatistics, Statistical Analysis, A/B Testing

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, K-Means, SMOTE

Deep Learning: TensorFlow, Keras, CNNs, Image Processing, Data Augmentation

Data Engineering: ETL/ELT, MS SQL Server, SQLAlchemy, PyODBC, SQLite, Star Schema

AI & Tools: LangGraph, LLMs, SHAP, Streamlit, Gradio, Docker, Git/GitHub

PROFESSIONAL EXPERIENCE

Data Science Intern — Madras Diabetes Research Foundation (MDRF) *Mar 2026 – Apr 2026; Jun 2026 – Jul 2026*

- Worked on healthcare data engineering and analytics workflows using Microsoft SQL Server, T-SQL, Python, and Pandas.
- Built a clinical data-cleaning and drug-standardization pipeline that resolved 15+ medication naming variations, removed approximately 12% duplicate records, and produced complete key fields in the final dataset.
- Analyzed longitudinal diabetes records for 5,172 patients across 48,864 follow-up records using FBS, PPBS, HbA1c, and OGTT-based clinical rules to identify disease progression and calculate progression timelines.
- Designed a modular healthcare ETL pipeline using Python, Pandas, SQLAlchemy, and PyODBC, including wide-to-long transformation, schema validation, MRNO masking, full-load, and incremental processing.

PROJECTS

AI-Based Liver Tumor Detection & Diagnostics | *Python, TensorFlow, Keras, Computer Vision*

- Built an end-to-end deep learning pipeline for liver tumor classification from abdominal CT scans, covering preprocessing, model development, and deployment.
- Applied spatial normalization, contrast optimization, and class-balancing techniques; deployed the model as an interactive application on Hugging Face Spaces.

FraudGuard AI — Insurance Claims Intelligence | *XGBoost, LightGBM, CatBoost, SHAP, Gradio*

- Developed an ensemble fraud-detection pipeline achieving 0.8603 ROC-AUC and 85.88% recall using 5-fold stratified cross-validation.
- Engineered interaction features and applied SMOTE within training folds to prevent data leakage; integrated SHAP for interpretable risk explanations.

Enterprise Population Health & Risk Stratification Portal | *Python, Random Forest, SHAP, Streamlit*

- Developed a leak-proof risk prediction workflow using GroupShuffleSplit across 99,343 real-world records to reduce patient and record leakage.
- Applied cohort eligibility rules, high-recall model tuning, and SHAP-based explanations; optimized the Streamlit application using caching.

Multi-Agent Clinical Trial Matcher | *LangGraph, Groq/Llama-3.3, Pydantic v2, Streamlit, ClinicalTrials.gov API v2*

- Built a multi-agent AI workflow that parses unstructured case notes and coordinates specialized agents to retrieve and score clinical trial matches.

- Implemented schema validation with Pydantic and integrated the ClinicalTrials.gov API to ground outputs in live registry data.

EDUCATION

M.Sc. Health Data Science (Final Year)

Expected 2027

SRM Institute of Science and Technology, Chennai, Tamil Nadu

B.Sc. Anaesthesia and Operation Technology

Narayana Hrudayalaya Institute of Health and Science, Bangalore, Karnataka

CERTIFICATIONS

- Image Processing Using Deep Learning
- Machine Learning & Deep Learning
- SQL Database Management & Python Programming
- SDG 3 Sustainable Development Goals
- SDG 9 Hackathon Participant

LANGUAGES

English, Kannada, Tamil, Telugu